FFGFT Field Theory: Comprehensive Presentation

Torus modes, Lagrangian, mode operator

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Part I
Foundations and Motivation

1 What is FFGFT?

FFGFT (Fundamental Fractal Geometric Field Theory), also called T0 Time-Mass Duality, describes all known elementary particles as bound oscillation modes of a **universal fractal torus**. The central idea: observed masses and interactions follow directly from geometry, without arbitrary parameters.

The theory requires only two fundamental inputs:

- **The geometric constant** $\xi_0 = \frac{4}{3} \times 10^{-4}$ — not postulated but derived from the tetrahedral symmetry of the 4D torus and the self-consistency of the electron-proton-muon system (Doc. 180).
- **The Planck length** ℓ_P — the natural length scale of quantum gravity.

All physical quantities — particle masses, coupling constants, cosmological parameters — follow from these two quantities.

2 The Fundamental Scale of the Torus

The characteristic length scale of the universal torus:

$$L_0 = \frac{\ell_P}{\xi_0} \approx 7500 \cdot \ell_P \approx 6.14 \times 10^{-16}\,\text{GeV}^{-1} \tag{1}$$

This is the **fundamental period** of the universal torus. All particles are modes on this torus — like overtones of a vibrating string, but in three spatial dimensions. The

Compton wavelength $L_i = 2\pi/m_i$ is the wavelength of the respective mode, not the size of a separate torus.

3 The Fractal Dimension

An ordinary torus has dimension 3. The FFGFT torus has a **fractal correction**:

$$D_f = 3 - \xi_0 \approx 2.9998667 \tag{2}$$

This minimal deviation is the origin of all particle masses. It generates a **non-linear dispersion relation** — analogous to an anharmonic crystal lattice. On a simple torus ($D_f = 3$) masses would be proportional to wave number: $m_n \propto n$, ratios integer. In reality:

$$\frac{m_\mu}{m_e} \approx 206.8, \quad \frac{m_\tau}{m_e} \approx 3477$$

These non-integer ratios require the fractal dispersion.

Part II

The Torus Modes and Their Quantum Numbers

4 Quantum Numbers of the Modes

Each mode on the torus is characterised by three quantum numbers: n (principal), l (orbital angular momentum, transverse component) and j (spin, $j = l \pm \frac{1}{2}$ for fermions).

The wave vector of a mode is quantised:

$$\mathbf{k}_{n,l} = \frac{2\pi}{L_0}(k_x, k_y, k_z), \quad k_x, k_y, k_z \in \mathbb{Z} \tag{3}$$

Connection to the quantum numbers:

$$n = k_x^2 + k_y^2 + k_z^2 \tag{4}$$

(principal = squared total length)

$$l = \sqrt{k_x^2 + k_y^2} \tag{5}$$

(orbital = transverse component)

The r_i prefactors and p_i exponents are discrete winding ratios of the 4D torus — not continuous parameters. For the leptons:

Particle	**n**	**l**	(k_x, k_y, k_z)	r_i	p_i
Electron	1	0	$(1, 0, 0)$	$4/3$	$3/2$
Muon	2	1	$(1, 1, 0)$	$16/5$	1
Tau	3	2	$(1, 1, 1)$	$25/9$	$2/3$

These values are the complete enumeration of discrete torus modes — analogous to the integers in hydrogen, where one does not ask for a formula for $n = 1, 2, 3$. The table (Doc. 006) is the theory. **Caveat:** this "table is the theory" stance is binding for practice. It does *not* imply the absence of structural depth; see §22 for the current status: lepton exponents form a geometric sequence with ratio $2/3$, lepton r_i carry an α-π-$\sqrt{3}$ deep structure (Doc. 070 §25), while the top-quark special case, the down-type step size, and the quark r_i structure remain open.

5 Explicit Mode Functions

On the torus $T^3 \times \mathbb{R}$, the eigenfunctions of the d'Alembertian $\Box = \partial_t^2 - \nabla^2$ are:

$$\boxed{\phi_{n,l,j}(t, \mathbf{x}) = \frac{1}{\sqrt{V}}\, e^{i\mathbf{k}_{n,l}\cdot\mathbf{x}} \cdot Y_{l,j}(\theta, \varphi) \cdot e^{-i m_{n,l,j}\, t}} \tag{6}$$

with $V = L_0^3$ (torus volume) and spherical harmonics $Y_{l,j}(\theta, \varphi)$.

These functions are **orthonormal**:

$$\int_{T^3} \phi^*_{n,l,j}(\mathbf{x})\, \phi_{n',l',j'}(\mathbf{x})\, d^3x = \delta_{nn'}\, \delta_{ll'}\, \delta_{jj'} \tag{7}$$

and satisfy the **Klein-Gordon equation**:

$$\Box\, \phi_{n,l,j} = -m^2_{n,l,j}\, \phi_{n,l,j} \tag{8}$$

The spin structure follows from the winding numbers of the torus (Doc. 051): spin-s corresponds to torus winding (n_θ, n_ϕ) with $n_\phi/n_\theta = 2s$. For spin-$\frac{1}{2}$: $(1, 1)$; the 720° symmetry follows directly from the topology.

Part III

The Non-Closure Theorem

6 Algebraic Structure of the Modes

The particle masses follow the geometric mass formula:

$$m_{n,l,j} = r_{n,l,j} \cdot \xi_0^{p_{n,l,j}} \cdot v, \qquad v = 246\,\text{GeV} \tag{9}$$

The prefactors r_i are rational numbers (labels of torus modes); the exponents p_i are rational scale parameters.

Restriction: Numerical substitution is permitted. Symbolic cross-multiplication is forbidden.

7 The Theorem

Non-Closure Theorem of FFGFT Torus Modes

For all pairs of different torus modes $i \neq j$:

$$\forall\, i \neq j : \; (r_i \cdot r_j,\; p_i + p_j) \notin \mathcal{P}$$

where $\mathcal{P}$ is the set of all admissible pairs (r, p) in FFGFT. The set $\mathcal{P}$ is **not closed** under cross-operations between different torus modes.

Proof of violation (example):

$$r_e \cdot r_\mu = \frac{4}{3} \cdot \frac{16}{5} = \frac{64}{15} \approx 4.267$$
$$p_e + p_\mu = \frac{3}{2} + 1 = \frac{5}{2}$$

In the complete mode table (Doc. 006), no particle has $r = 64/15$ and $p = 5/2$. The pair $(64/15, 5/2) \notin \mathcal{P}$.

Why does the theorem hold? The cause lies in the orthogonality of the modes. In Hilbert space, the product of two different basis vectors is structurally meaningless. Likewise, $r_e \cdot r_\mu$ is computable as a number but belongs to no torus mode.

Practical rule:

Permitted	**Forbidden**
Numerical: $m_e \cdot m_\mu \approx 53.9$ MeV^2	Symbolic: $r_e \cdot r_\mu = 64/15$ as mass formula
$m_i = r_i \cdot \xi_0^{p_i} \cdot v$ for fixed i	$(r_i \cdot r_j) \cdot \xi_0^{p_i+p_j} \cdot v$ for $i \neq j$
Mass ratios m_i/m_j numerically	Cross-multiplication of different r_i, r_j

Part IV

The Lagrangian and the Bridge to the Mode Operator

8 The Free Lagrangian

Starting point is the free Klein-Gordon Lagrangian:

$$\mathcal{L}_0 = \frac{1}{2}(\partial \delta m)^2 \tag{10}$$

The scalar field $\delta m(x)$ describes mass fluctuations on the torus. The equation of motion $\Box \delta m = 0$ describes massless oscillations.

Related document: Doc. 201 (DVFT adaptation)

Doc. 201 (*FFGFT-alles*) develops the same starting point $\mathcal{L}_0 = \varepsilon(\partial\Delta m)^2$ in a different direction: the polar decomposition $\Phi = \rho e^{i\theta}$ with vacuum amplitude ρ and vacuum phase θ (DVFT — Dynamic Vacuum Field Theory). Applications: *apparent* cosmic acceleration (without dark energy), galactic rotation curves (without dark matter), CMB homogeneity. No contradiction with Doc. 202 — different development directions of the same foundation.

Quantum-mechanical equations of motion: The DVFT line in Doc. 201 §2.6 (simplified Dirac equation from T0) and §6.6 (Schrödinger equation derivation from T0) provides the bridge to non-relativistic and relativistic quantum mechanics. In the mode picture of Doc. 202, the DVFT polar decomposition corresponds to the low-energy limit of the mode superposition; the explicit bridge formula is documented in the section “Connection to the Quantum Mechanical Equations of Motion” below.

Note on the terminology “renormalisation group”: The scale running of the coupling in Doc. 202 § 18 is referred to there – following standard QFT language – as the *renormalisation group*. In FFGFT, however, this scale structure follows directly from the recursion $\mathcal{R}$ (Doc. 203) on the fractal torus geometry and is *not* the result of a renormalisation procedure. Doc. 201 consequently avoids this terminology and describes the scale flow directly through the DVFT phase dynamics (ρ, θ) – i.e. in an FFGFT-native manner. The unified language convention (scale structure from recursion, $\mathcal{R}$-induced scale correction instead of RG language) is recorded in Doc. 190 R7 and is binding for the entire series.

9 The Torus Correction Term

The fractal geometry adds a correction term generating the non-linear dispersion:

$$\Delta\mathcal{L}_{\text{Torus}} = \varepsilon\,\xi_0\left[\frac{f(\theta)}{2}(\partial\delta m)^2 - k^{\mu\nu}(\theta)\,\partial_\mu\delta m\,\partial_\nu\delta m\right] \tag{11}$$

The geometric functions $f(\theta)$ and $k^{\mu\nu}(\theta)$ encode the curvature of the fractal torus. The complete FFGFT Lagrangian:

$$\boxed{\mathcal{L} = \mathcal{L}_0 + \Delta\mathcal{L}_{\text{Torus}}} \tag{12}$$

10 The Mode Operator

The Hilbert space of torus modes:

$$\mathcal{H} = \bigoplus_{n,l,j}\mathbb{C}\cdot\mathbf{e}_{n,l,j}, \qquad \langle\mathbf{e}_{n,l,j},\mathbf{e}_{n',l',j'}\rangle = \delta_{nn'}\delta_{ll'}\delta_{jj'} \tag{13}$$

The field operator:

$$\Phi(x) = \sum_{n,l,j} m_{n,l,j}\cdot\phi_{n,l,j}(x)\cdot P_{n,l,j} \tag{14}$$

with $P_{n,l,j} = |\mathbf{e}_{n,l,j}\rangle\langle\mathbf{e}_{n,l,j}|$. Since $P_iP_j = \delta_{ij}P_i$: $\Phi^2 = \sum_i m_i^2 P_i$.

11 The Bridge Formula

The Lagrangian and mode operator describe the same physics:

$$\boxed{\int\frac{1}{2}(\partial\delta m)^2\,d^4x = \frac{1}{2}\langle\psi|\Phi^2|\psi\rangle} \tag{15}$$

Derivation in 6 steps:

Step 1 — Mode expansion: $\delta m(x) = \sum_i a_i\phi_i(x)$.

Step 2 — Substitution: $\int\mathcal{L}_0\,d^4x = \frac{1}{2}\sum_{i,j}a_ia_j\int(\partial\phi_i)(\partial\phi_j)\,d^4x$.

Step 3 — Orthogonality: Cross terms vanish for $i \neq j$, leaving $\frac{1}{2}\sum_i a_i^2\int(\partial\phi_i)^2\,d^4x$.

Step 4 — Klein-Gordon: $\int (\partial\phi_i)^2\, d^4x = m_i^2$ (partial integration, normalisation).

Step 5 — Summary: $\int \mathcal{L}_0\, d^4x = \frac{1}{2}\sum_i a_i^2 m_i^2$.

Step 6 — Mode operator: $\langle\psi|\Phi^2|\psi\rangle = \sum_i a_i^2 m_i^2$. Comparison gives (15) with $\varepsilon = 1/2$.

Part V

Mass Generation as a Binding Phenomenon

12 The Binding Picture

The eigenvalue equation on the torus:

$$\left[\Box + \hat{V}(\xi_0, D_f)\right] \phi_{n,l,j} = -m_{n,l,j}^2\, \phi_{n,l,j} \tag{16}$$

The mass is a difference:

$$m_i^2 = \underbrace{\left(\frac{2\pi n}{L_0}\right)^2}_{\text{kinetic}} + \underbrace{\langle\phi_i|\hat{V}|\phi_i\rangle}_{\approx -(2\pi n/L_0)^2} \tag{17}$$

The kinetic energy is enormous ($\sim 10^{32}$ GeV^2), the potential almost exactly opposite. The mass is the tiny residual:

Particle	$(2\pi n/L_0)^2$	$m_i^2/$**(kin.)**
Electron	1.046×10^{32} GeV^2	2.5×10^{-39}
Muon	4.185×10^{32} GeV^2	2.7×10^{-35}
Tau	9.415×10^{32} GeV^2	3.4×10^{-33}

13 Not Fine-Tuning — a Binding Phenomenon

The extreme cancellation (32–39 orders of magnitude) is not fine-tuning — it is the defining property of bound states.

Crystal analogy: One does not ask why a crystal produces a particular phonon dispersion. The dispersion curve *is* the crystal structure in momentum space.

Hydrogen analogy: Kinetic energy $\approx$ $13.6\,$eV, Coulomb potential $\approx -27.2\,$eV, binding energy $-13.6\,$eV. Nobody asks why the potential compensates the kinetic energy so precisely — it follows from the Schrödinger equation.

In FFGFT: The particles are bound torus modes. The mass is the binding residual. The potential $\hat{V}$ is uniquely determined by the torus geometry (ξ_0, D_f) — no free parameter.

14 Connection to the Lagrangian Extension

Lagrangian formulation	Torus binding picture
$\xi_0 \cdot m_\ell \cdot \bar{\psi}\psi \cdot \delta m$ (coupling)	Coupling of fermion mode to its own torus mode
$\Delta\mathcal{L}_{\text{Torus}}$	Potential term $\hat{V}$ from $D_f = 3 - \xi_0$
Mass parameter m_ℓ appears explicitly	Mass emerges as eigenvalue residual
Standard + geometric correction	Kinetic $+$ potential ≈ 0, residual $= m_i^2$

Difference from the Higgs mechanism: In the Standard Model, the Yukawa coupling y_i is a free parameter. In FFGFT, the corresponding term $\xi_0 \cdot m_\ell$ is geometrically fixed by $\xi_0 = 4/30000$.

The central claim:

Standard QFT calculations must be extended by a tiny geometric mass contribution that arises directly from the torus structure.

15 Connection to the Quantum Mechanical Equations of Motion

The mode structure developed in Doc. 202 and the free Lagrangian $\mathcal{L}_0 = \frac{1}{2}(\partial\delta m)^2$ provide the *spectral framework*: masses arise as eigenvalues of the mode operator, not as Lagrangian parameters. For the quantum-mechanical equations of motion the parallel DVFT line in Doc. 201 is invoked, which develops the same starting point $\mathcal{L}_0 = \varepsilon(\partial\Delta m)^2$ in a different direction.

Bridge 202 $\leftrightarrow$ 201

In the low-energy limit, the mode sum of Doc. 202 maps to the DVFT polar decomposition of Doc. 201:

$$\Phi(x) \;=\; \sum_{n,l,j} m_{n,l,j}\,\phi_{n,l,j}(x)\,P_{n,l,j} \quad \xrightarrow{\text{low-energy}} \quad \sqrt{\rho(x,t)}\,e^{i\theta(x,t)} \tag{202.QM-1}$$

with $\rho(x,t) = \left|\sum_{n,l,j}\alpha_{n,l,j}\phi_{n,l,j}(x)\right|^2$ as energy density and $\theta(x,t)$ as the coherent phase.

Comprehensive sources for the QM extension: Doc. 020 (T0-QM-QFT-RT, comprehensive treatment), Doc. 035 (QM overview).

Simplified Dirac equation (from Doc. 201 § 2.6)

On this basis, Doc. 201 replaces the full Dirac equation by:

$$\partial^2\,\Delta m \;=\; 0 \tag{202.QM-2}$$

The fundamental structure here is the Clifford algebra $\mathbf{e}_\mu\mathbf{e}_\nu + \mathbf{e}_\nu\mathbf{e}_\mu = 2g_{\mu\nu}$ (Doc. 050, 051) rather than a specific matrix representation. The 4×4 γ matrices nevertheless remain *as a representation* of the Clifford generators and stay a practical tool for concrete calculations — they are therefore not eliminated but rather traced back to their geometric meaning. Spin manifests itself on the fractal torus as a topological property (*vacuum phase winding*), consistent with the quantum number $j = l \pm \frac{1}{2}$ from the mode catalogue of Doc. 202 §"Quantum Numbers of the Modes". A relativistic state becomes

$$\Phi(x,t) \;=\; \rho(x,t)\, e^{i\theta(x,t)} \cdot \chi(\sigma) \tag{202. QM-3}$$

with $\chi(\sigma)$ as the spin component arising from geometric knot modes.

Parallel derivations on the Dirac line: Doc. 020 §"T0-modified Dirac equation", Doc. 067 §"Modified Dirac equation", Doc. 095 §"Simplified Dirac equation (T0 version)", Doc. 053 (mass elimination in the Dirac Lagrangian).

Schrödinger equation (from Doc. 201 §6.6)

In the non-relativistic limit, the phase equation of the vacuum field reduces to the standard Schrödinger equation:

$$i\hbar\,\frac{\partial\psi}{\partial t} \;=\; -\frac{\hbar^2}{2m}\,\nabla^2\psi + V\psi \tag{202. QM-4}$$

with $\psi \;\propto\; \sqrt{\rho}\, e^{i\theta}$, effective mass m from the knot inertia, and potential V from external ρ perturbations. The Schrödinger equation is therefore *not fundamental* but an effective equation for slow, non-relativistic knot excitations.

Parallel derivations on the Schrödinger line: Doc. 020 §"T0-modified Schrödinger equation", Doc. 067 §"Modified Schrödinger equation", Doc. 095 §"Extended Schrödinger equation (T0-modified)", Doc. 129 §"Schrödinger equation in simplified T0 form", Doc. 037 (Cairo counterexample and geometric resolution).

Bell correlations (from Doc. 023, 074)

Entangled states correspond in FFGFT to correlated mode configurations sharing a common phase winding on the torus. The modified CHSH form contains geometric constants that follow directly from the Lagrangian framework:

$$\mathrm{CHSH}^{\mathrm{FFGFT}} = 2\sqrt{2} \cdot K_{\mathrm{frak}}^{D_f} \cdot \left(1 - \xi_0 \cdot \frac{\Delta\theta}{\pi}\right) \qquad (202.\,\mathrm{QM\text{-}5})$$

with $D_f = 3 - \xi_0$ (fractal dimension from the torus correction term $\Delta\mathcal{L}_{\mathrm{Torus}}$, § 9), K_{frak} as geometric correction factor, and $\xi_0 = 4/30000$. The prediction is a tiny deviation from the standard quantum value:

$$\Delta\mathrm{CHSH} = 2\sqrt{2} - \mathrm{CHSH}^{\mathrm{FFGFT}} \sim \mathcal{O}(\xi_0) \approx 10^{-4} \qquad (202.\,\mathrm{QM\text{-}6})$$

Lagrangian connection. Bell correlations do not follow directly from $\mathcal{L} = \mathcal{L}_0 + \Delta\mathcal{L}_{\mathrm{Torus}}$, but from the *topology* of the mode configurations. The Lagrangian nevertheless supplies the constants ξ_0, D_f and K_{frak} appearing in (202. QM-5). Bell tests thereby become an *indirect* probe of the Lagrangian structure via the ξ_0 scale constant.

Locality picture. The correlations arise from topological knot correlation on the torus, not from non-local action. Entanglement appears as a shared phase winding between two mode sectors with correlated quantum numbers (n, l, j). Experimentally accessible in loophole-free tests at large angles $\Delta\theta > \pi$ with resolution $\sim 10^{-4}$ (Doc. 023, 074, 131).

Parallel derivations on the Bell line:

- Dedicated Bell series: Doc. 023 (Bell tests in the T0 context, modified CHSH form), Doc. 023a Bell-Part 2 (multi-qubit extension, philosophical reflections, experimental proposals), Doc. 023a Bell-Video (locally realistic picture in detail).

- Quantum computing application: Doc. 147 (Bell test modifications §, 73-qubit predictions ΔCHSH $\sim 10^{-3}$, Bell-enhanced peak detection).
- Locality & non-locality: Doc. 074 (no-go theorem and T0 answer to Bell), Doc. 131 ("apparently instantaneous" – locality picture), Doc. 055 (DynMassePhotons non-local).
- Cross-cutting: Doc. 022 (T0-QFT-ML addendum), Doc. 035 § "Bell tests & EPR", Doc. 175 (qubit state spaces).

Qubit mappings (from Doc. 175, 034)

A qubit state is, in the FFGFT picture, a specific configuration of the mode structure. From the bridge formula (202. QM-1) one obtains two localised basis configurations:

$$\Phi_0(x) \;=\; \rho_0(x)\, e^{i\theta_0(x,t)}, \qquad \Phi_1(x) \;=\; \rho_1(x)\, e^{i\theta_1(x,t)}. \tag{202. QM-7}$$

The coherent superposition $\alpha\Phi_0+\beta\Phi_1$ ($|\alpha|^2+|\beta|^2=1$) yields the general qubit state as a *single phase configuration* of the vacuum field — not an ontological multiple existence but a coherent superposition of modes.

Cylindrical phase space. The natural geometry of a qubit in FFGFT (Doc. 034, 175) is not the Bloch sphere but a cylindrical phase space with three coordinates (z, r, θ):

- $z \in [-1, +1]$: torsion configuration along $|0\rangle \leftrightarrow |1\rangle$
- $r \in [0, 1]$: strength of intermediate torsion
- $\theta \in [0, 2\pi)$: relative phase

with the Bloch-sphere correspondence $z = \cos\theta_{\text{Bloch}}$, $r = |\sin\theta_{\text{Bloch}}|$. The consequence: *quantum gates become geometric transformations in real space* — the Hadamard gate exchanges $z \leftrightarrow r$ with a $\pi/2$ phase rotation; the Z phase is a pure rotation about the cylinder axis.

Lagrangian connection. The basis states $|0\rangle, |1\rangle$ are two specific mode excitations $\phi_{n,l,j}$ from the mode catalogue of Doc. 202 §"Quantum Numbers of the Modes" and §"Explicit Mode Functions". The geometry of the mode functions arises from the Lagrangian $\mathcal{L} = \mathcal{L}_0 + \Delta\mathcal{L}_{\text{Torus}}$; qubit operations are effective transformations on the Hilbert space spanned by the mode structure. Spin qubits sit at the ξ level $N \approx 8$ (nanometre scale, quantum confinement); photonic resonators sit on a different level of the ξ hierarchy (Doc. 175).

Parallel derivations on the qubit line:

- Geometry and phase space: Doc. 034 (QM optimisation, introduction of the cylindrical phase space), Doc. 175 (qubit state spaces, quantum registers), Doc. 173 (quantum-dot resonator, level $N \approx 8$).
- Spin manipulation and readout: Doc. 174 (five methods for spin manipulation, QND measurement), Doc. 178 (FFGFT spin stability).
- Quantum computing architecture: Doc. 147 (ϕ-hierarchical QFT, Shor's algorithm, deterministic energy-field qubits), Doc. 176 (Shor-photon), Doc. 161 (Ising machine), Doc. 183 (Willow self-measurement).

Consequence for the Verification Condition

Both equations of motion (202. QM-2) and (202. QM-4) are contained in the mode picture of Doc. 202 – as different low-energy limits of the same spectral structure. The verification condition of the next section therefore applies consistently to both sectors. The Bell correlation (202. QM-5) and the qubit mapping (202. QM-7) provide independent consistency tests of the same ξ_0 constant and the same mode structure.

16 The Verification Condition

The theory makes a numerically testable prediction:

$$\boxed{\langle\phi_{n,l,j}|\hat{V}|\phi_{n,l,j}\rangle = (r_i)^2 \cdot \xi_0^{2p_i} \cdot v^2 - \left(\frac{2\pi n}{L_0}\right)^2} \tag{18}$$

This is the Stage 3 bottleneck (Doc. 189): explicitly derive $\Delta\mathcal{L}_{\text{Torus}}$ from the torus geometry and show that the eigenvalues match the mass table (Doc. 006).

17 Methodological Status of the Verification Condition

The verification condition (18) is evaluated against experimental masses. This evaluation carries three methodological limitations that must be made explicit, since they are often passed over.

Reduction is not precision improvement

The FFGFT Lagrangian $\mathcal{L} = \mathcal{L}_0 + \Delta\mathcal{L}_{\text{Torus}} + \mathcal{L}_{\text{Yukawa}}$ contains the Standard-Model Lagrangian as a limiting case: for $\xi_0 \to 0$ both $\Delta\mathcal{L}_{\text{Torus}}$ and the Yukawa term $\xi_0 \cdot m_\ell \cdot \bar{\psi}\psi \cdot \delta m$ vanish. Where standard calculations are experimentally confirmed (e.g. QED scattering amplitudes to $\sim 10^{-12}$), FFGFT can only reproduce these and add ξ_0 corrections of order 10^{-4}, not surpass them qualitatively.

The FFGFT mass formulas $m_\ell = r_\ell \cdot \xi_0^{p_\ell} \cdot v$ yield values agreeing with PDG values to $\sim 1\,\%$ (Doc. 190 C5). This is *not* a precision result but a *structural proof*: the masses follow from the geometry, not from free parameters. The reduction of ~ 25 free Standard-Model parameters to a single geometric parameter ξ_0 is the quantifiable claim — not the numerical agreement of the individual masses.

Predictions beyond the Standard Model

Where FFGFT delivers *different* statements, these are predictions for phenomena where the Standard Model is either silent or predicts corrections not yet experimentally resolved:

- Bell CHSH correction ΔCHSH $\sim \mathcal{O}(\xi_0) \approx 10^{-4}$ in loophole-free tests at large angles (Doc. 023, 074, 147; see § 15).
- ΔCHSH$\sim$ 10^{-3} in 73-qubit systems with Bell-enhanced peak detection (Doc. 147).
- Modified dispersion at near-Planck energies (Doc. 055, 074).
- Dark matter and dark energy without separate components (DVFT line, Doc. 201).
- CMB homogeneity without inflation (Doc. 140, 201).
- Hierarchy problem: Higgs mass vs. Planck mass as a geometric consequence, not a fine-tuning problem (Doc. 003, 049).

Circularity of the comparison basis

A systematic limitation of any numerical competition with the Standard Model concerns the experimental comparison basis itself: PDG values are not model-free raw data but already contain assumptions about

- units and gauge conventions,
- calibrating natural constants (c, $\hbar$, e were fixed by definition in the 2019 SI reform),
- loop corrections from QFT evaluations.

These points are systematically treated in Doc. 066 (parameter-transfer problem between unit systems) and Doc. 101 (circularity of the constants).

An FFGFT calculation with a simpler entry point — starting directly from ξ_0 in natural units ($\hbar = c = 1$) — avoids the accumulation of model assumptions along the standard calibration chain. As soon as it measures its values against PDG data, however, it inherits their precorrections. The agreement at the $\sim 1\,\%$ level should therefore be read in two directions: as confirmation of the geometric structural claim *and* as a pointer to the latitude introduced by the circular measurement basis. A direct numerical competition would be fair only if the entire measurement chain were conceptually reorganised — a programme outlined in Doc. 066 and 101 but exceeding the scope of the present field-theoretic formulation.

Consequence for the assessment

The verification condition (18) is therefore to be read *structurally*, not *quantitatively*: if the eigenvalues of the mode-operator equation are confirmed to match the mass table within the $\sim 1\,\%$ band, the geometric derivation is supported. The remaining margin is not to be interpreted as a theoretical deficit but as a reflex of the three methodological limitations stated above.

Part VI
The Field-Theoretic Formulation

18 Hilbert Space and Recursion Operator

The recursion operator $\mathscr{R}$ formalises the fractal scaling structure:

$$\mathscr{R}(\mathbf{e}_{n,l,j}) = \xi_0^{p(n,l,j)-1} \cdot K_{\mathsf{frak}} \cdot \mathbf{e}_{n,l,j}, \qquad \mathscr{R}(\Phi_*) = \Phi_* \tag{19}$$

The time-mass duality is a field symmetry: $\mathscr{R}: \quad t \to \xi_0 t,\ m \to \xi_0^{-1} m,\ \Phi \to \Phi$.

19 Algebraic Structure

Mode algebra for free waves on the torus:

$$\mathbf{e}_{\mathbf{k}_i} \star \mathbf{e}_{\mathbf{k}_j} = \mathbf{e}_{\mathbf{k}_i+\mathbf{k}_j}, \qquad C_{ij}^{k} = \delta_{k,\,i+j} \quad (\text{mod torus period}) \tag{20}$$

20 Scale Structure from Recursion

The recursive α running in FFGFT arises from the iterated application of the recursion operator $\mathscr{R}$ (Doc. 203) to the mode structure. In the formal writing of a modified QED β-function (Doc. 008):

$$\frac{d\alpha}{d\ln\mu} = \beta_0\alpha^2 \cdot \left(1 + \xi_0 \cdot \ln\frac{\mu}{E_0}\right) \tag{21}$$

$$\alpha^{-1}(\mu) = \alpha^{-1}(m_e) - \frac{\beta_0}{2\pi}\ln\frac{\mu}{m_e} - \frac{\beta_0\xi_0}{4\pi}\left(\ln\frac{\mu}{m_e}\right)^2 \tag{22}$$

The ξ_0 term is the FFGFT-specific addition. ξ_0 is **not** a fixed point of this β-function but a fundamental geometric constant — time-invariant because topologically fixed.

Note on terminology. The scale running was occasionally referred to as the "renormalisation group" in earlier FFGFT documents. This designation is inaccurate: the scale flow here is *not* the result of a renormalisation procedure but a direct consequence of the recursion $\mathscr{R}$ on the fractal torus geometry. Stability, scale flow and UV finiteness are three aspects of the same recursive structure. The binding language convention is recorded in Doc. 190 R7; the FFGFT-native view is already developed in Doc. 159 ("without artificial renormalisation") and Doc. 189 ("geometrically determined, not by renormalisation").

Part VII

Open Mathematical Tasks

21 Four Precisely Formulated Problems

The theory is conceptually closed. The remaining tasks are mathematical.

Problems 1 and 2 have been formally worked out in Doc. 203 ($\mathscr{R}$-operator, fixed-point condition) and Doc. 204 (fractal boundary condition, first-order perturbation in ξ_0). Problem 3 (algebra) is complete for the free theory. Problem 4 (interactions) remains open.

No.	Problem	Task	Status
1	$\mathscr{R}$-operator	$\lambda_{n,l,j} = \xi_0^{p-1} K_{\text{frak}}$; $p(n,l,j)$ from torus	Doc. 203
2	Mode functions	Incorporate fractal boundary condition	Doc. 204
3	Algebra	Hopf algebra for interactions	Free theory: complete
4	Interactions	$\mathscr{L}_{\text{int}}$, β-function	Doc. 008; ξ_0 geometric

22 Status of the (r_i, p_i) structure

The (r_i, p_i) values tabulated in Doc. 006 reproduce all twelve Standard Model fermion masses to $< 2\,\%$ accuracy (parameter-free, $\xi_0 = 4/30000$, $v = 246\,$GeV). The “hydrogen argument” (§4 of this document) – the table is the theory – remains binding for practice. This section makes explicit what is structurally understood and what is not.

22.1 What is structurally understood

Three generations. The $\mathbb{Z}_3$-grading follows from the torus topology and is topologically grounded (Doc. 172, Avi-Rosenthal dialogue: $27 = 3^3$ from $SU(27) \supset SU(9)^3 \supset SU(3)^9 \supset SU(3) \times SU(2) \times U(1)$).

Lepton exponents form a geometric sequence. The values $(p_e, p_\mu, p_\tau) = (3/2, 1, 2/3)$ satisfy

$$p_{n+1} = \tfrac{2}{3}\, p_n, \qquad \text{geometric sequence of ratio } \tfrac{2}{3}. \tag{23}$$

The factor $2/3$ is the inverse of the tetrahedral factor $3/2$ and directly linked to $\xi_0 = 4/30000 = 4/3 \cdot 10^{-4}$.

Up-type quarks follow the same scheme up to charm. Up and charm have $p_u = 3/2$ and $p_c = 2/3 = (2/3)^2 \cdot p_e$ – the lepton sequence with one stage skipped. With top, the scheme breaks (see below).

Lepton r_i admit a mode-local α structure statement. Doc. 070 §25 shows that the apparently simple fractions $4/3, 16/5, 25/9$ admit, *per mode*, decompositions in α, π, $\sqrt{3}$:

$$r_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \quad r_\mu = \frac{9}{4\pi\alpha}, \quad r_\tau = \frac{27\sqrt{3}}{8\pi\alpha^{3/2}}. \tag{24}$$

These are *three separate structural equations*, one per mode. A closed $\alpha(\xi_0)$ relation requires a unit convention, discussed in §22.3.

22.2 What remains open

Top-quark special case. Top has $p_t = -1/3$ (negative sign!) and $r_t = 1/28$. The lepton/up-quark sequence with factor $2/3$ would predict $p_t = (2/3)^3 \cdot p_e = 4/9 \approx 0.44$, not $-1/3$. The jump $p_c = 2/3 \to p_t = -1/3$ corresponds to factor $-1/2$, without a known geometric basis. Doc. 006 labels this “inverted hierarchy” without derivation.

Down-type quarks follow a different scheme. $(p_d, p_s, p_b) = (3/2, 1, 1/2)$ is an *arithmetic* sequence of constant step $\Delta p = -1/2$, not the geometric sequence with factor $2/3$ of the leptons. Why up-type and down-type follow different sequence types is unexplained.

Quark r_i values without α structure. While lepton r_i are decomposed in Doc. 070 §25 in α-π-$\sqrt{3}$, the analogous depth structure is missing for the quark values $\{6, 25/2, 26/9, 2, 1/28, 3/2\}$. Doc. 006 labels them only with "color" or "color + isospin".

Complete Hopf algebra for interactions. For the free theory, $C_{ij}^k = \delta_{k,i+j}$ (wave-vector addition). The extension to interaction terms with fully worked-out structure constants and Feynman rules remains.

22.3 "Maximal formula": unit convention and use

In Doc. 070 §3.6 and §16 a closed form for α in terms of ξ_0 appears:

$$m_e = c_e \cdot \xi_0^{5/2}, \qquad m_\mu = c_\mu \cdot \xi_0^2 \tag{070-4.2}$$

$$E_0 = \sqrt{m_e \cdot m_\mu} = \sqrt{c_e \cdot c_\mu} \cdot \xi_0^{9/4} \tag{070-3}$$

$$\alpha = \xi_0 \cdot E_0^2 \ = \ c_e \cdot c_\mu \cdot \xi_0^{11/2} \tag{070-3.6.0}$$

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi_0^{11/5} \cdot K_{\mathsf{frak}} \tag{070-3.6.1}$$

Yukawa form and maximal formula are the same theory in two notations, linked by the bridge formula of Doc. 210 W5:

$$m_e = r_e\, \xi_0^{p_e}\, v \iff m_e = c_e\, \xi_0^{5/2} \quad \text{with} \quad c_e = r_e \cdot v \cdot \xi_0^{p_e - 5/2}.$$

The apparent discrepancy between the two representations is a *question of unit convention*, not an inconsistency:

Unit convention of the maximal formula. The form (070-3.6.0) is exact provided all masses are expressed in a *theory-internal scale* S_{T0}. Doc. 013 (SI matching) postulates

$$S_{T0} \ = \ 1\,\text{MeV}/c^2 \tag{25}$$

as the geometrically fixed scaling factor. In this convention, $c_e = m_e/(\xi_0^{5/2}\, S_{T0})$ and $c_\mu = m_\mu/(\xi_0^2\, S_{T0})$ are *dimensionless* (numerically $\sim 10^9$), and (070-3.6.0) yields $\alpha \approx 7.2 \cdot 10^{-3}$, in agreement with experiment.

Connection via Higgs matching. Doc. 006 §EFT-Matching gives the fundamental relation

$$\xi_0 \;=\; \frac{\lambda_h^2\, v^2}{16\pi^3\, m_h^2} \tag{26}$$

with Higgs self-coupling $\lambda_h \approx 0.13$, Higgs VEV $v \approx 246$ GeV, Higgs mass $m_h \approx 125$ GeV. Hence v is *not* a free parameter but linked to ξ_0 via (26). Substituting into the transparent form $\alpha = r_e r_\mu \cdot v^2 \cdot \xi_0^{7/2}$ gives

$$\alpha \;=\; \frac{16\pi^3\, r_e\, r_\mu}{\lambda_h^2} \cdot m_h^2 \cdot \xi_0^{9/2}, \tag{27}$$

numerically $\alpha \approx 7.15 \cdot 10^{-3}$ ($< 2\,\%$ deviation).

Content of $K_{\textbf{frak}}$ in (070-3.6.1). In this convention,

$$K_{\text{frak}}^{\text{(maximal formula)}} \;=\; \frac{16\pi^3\, r_e\, r_\mu\, m_h^2}{\lambda_h^2\, (27\sqrt{3}/8\pi^2)^{2/5}} \cdot \xi_0^{9/2-11/5} \;\approx\; 3.0 \cdot 10^6\ \text{MeV}^2, \tag{28}$$

a *collective factor* absorbing the Higgs scale m_h^2/λ_h^2 and the ξ_0 exponent difference. Important: this $K_{\text{frak}}^{\text{(maximal formula)}}$ is **not** identical with the small geometric correction $K_{\text{frak}} = 1 - 100\xi_0 \approx 0.9867$ from Doc. 018; using the same symbol in both contexts is an ambiguity which future versions should resolve through distinct names (e.g. K_{geo} for 018, K_{scale} for 070-3.6.1).

Binding.

- The maximal formula (070-3.6.0) or (070-3.6.1) is *numerically usable* and reproduces α correctly, provided either (a) the scale convention $S_{T0} = 1$ MeV (Doc. 013) or (b) Higgs matching (26) (Doc. 006) is explicitly applied.

- Yukawa form and maximal formula are the same theory; their agreement is formally established by the bridge formula of Doc. 210 W5.
- The most transparent expression for communication is $\alpha = r_e r_\mu \cdot v^2 \cdot \xi_0^{7/2}$ or equivalently $\alpha = \xi_0 \cdot E_0^2$ with $E_0 = \sqrt{m_e m_\mu}$. When citing the maximal formula, the content of K_{frak} must be stated explicitly.
- Per mode, the structural identification $r_i = c_i \cdot \alpha^{q_i}$ with $q_e = -1/2, q_\mu = -1, q_\tau = -3/2$ (Doc. 070 §25.3.1) remains valid.

22.4 Recommendation for future communication

- The lepton exponents $(3/2, 1, 2/3)$ form a *geometric* sequence of ratio $2/3$, not an arithmetic sequence of step $1/3$.
- The claim "$\Delta p = 1/3 = 1/D$" (older versions of Doc. 202, Doc. 203 §3) is not literally satisfied; it holds only for the $p_\mu \to p_\tau$ step.
- Yukawa form (Doc. 006) and maximal formula (Doc. 070 §3.6) are the same theory; both are usable provided the scale convention ($S_{T0} = 1$ MeV from Doc. 013, or Higgs matching from Doc. 006) is explicitly named. When citing $\alpha = (27\sqrt{3}/8\pi^2)^{2/5}\xi_0^{11/5}K_{\text{frak}}$, the content of K_{frak} must be specified (it is *not* identical with the K_{frak} from Doc. 018).
- Top quark, down-type step size, and quark r_i depth structure are to be flagged explicitly as open items, rather than papered over by generalised schemata.

Part VIII

Summary

23 Overview of Results

FFGFT / T0 theory describes all known elementary particles as modes of a universal fractal torus — closed, consistent, from a single geometric constant $\xi_0 = 4/3 \times 10^{-4}$.

Concept	Meaning
Universal torus	$D_f = 3 - \xi_0$, fundamental scale $L_0 = \ell_P/\xi_0$
Torus modes	Quantum numbers (n, l, j), wave vector $\mathbf{k} = (2\pi/L_0)(k_x, k_y, k_z)$
Mode functions	$\phi = V^{-1/2} e^{i\mathbf{k}\cdot\mathbf{x}} Y_{l,j} e^{-imt}$
Lagrangian	$\mathcal{L} = \frac{1}{2}(\partial\delta m)^2 + \Delta\mathcal{L}_{\text{Torus}}$
Mode operator	$\Phi = \sum_i m_i P_i$, diagonal in Hilbert space
Bridge	$\int \mathcal{L}_0 \, d^4x = \frac{1}{2}\langle\psi|\Phi^2|\psi\rangle$
Mass formula	$m_i = r_i \cdot \xi_0^{p_i} \cdot v$, $r_i, p_i \in \mathbb{Q}$
Non-closure	$(r_i r_j, p_i + p_j) \notin \mathcal{P}$ for $i \neq j$
Mass generation	Bound states: kin. $\approx$ pot., mass $=$ binding residual

Notation

Symbol	Meaning
$\xi_0 = 4/30000$	Geometric base constant of the fractal torus
ℓ_P	Planck length
$L_0 = \ell_P/\xi_0$	Fundamental torus scale
$D_f = 3 - \xi_0$	Fractal dimension of the FFGFT torus
$v = 246\,\text{GeV}$	Higgs vacuum expectation value (energy unit)
$n,\ l,\ j$	Quantum numbers: principal, orbital, spin
$\mathbf{k}_{n,l}$	Quantised wave vector on the torus
$Y_{l,j}(\theta,\varphi)$	Spherical harmonics
$\phi_{n,l,j}(t,\mathbf{x})$	Torus mode functions
$m_i = r_i \cdot \xi_0^{p_i} \cdot v$	FFGFT mass formula
$r_i \in \mathbb{Q}$	Rational prefactor (winding ratio)
$p_i \in \mathbb{Q}$	Rational exponent (scaling depth)
$\mathcal{P}$	Set of admissible pairs (r_i, p_i)
$\mathcal{H}$	Hilbert space of torus modes
$\mathbf{e}_{n,l,j}$	Basis vector of the Hilbert space
$P_i = \lvert\mathbf{e}_i\rangle\langle\mathbf{e}_i\rvert$	Projector onto mode i
$\Phi = \sum_i m_i P_i$	Mode operator (field operator)
$\mathcal{L}_0 = \frac{1}{2}(\partial\delta m)^2$	Free Lagrangian
$\Delta\mathcal{L}_{\text{Torus}}$	Torus-geometric correction term
$\hat{V}(\xi_0, D_f)$	Potential operator from torus geometry
$K_{\text{frak}} \approx 0.986$	Fractal correction constant
$C_{ij}^k = \delta_{k,i+j}$	Structure constants of the mode algebra
β_0	Coefficient of the QED beta function

References: Doc. 006 (quantum numbers), Doc. 008 (β-function), Doc. 049 (L_0), Doc. 051 (spinor), Doc. 129, Doc. 180 (ξ_0), Doc. 189 (stage structure).